

Math 571 - Homework 3

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Problems numbered [B:2.1:3] are from [Bergman's exercise notes](#). Problems labeled [R:2.1] are directly from Rudin. If I add an asterisk (*) to a problem, we have a slight variant of the problem.

Notation: $\{p_i\} = \{p_i \mid i = 1, 2, \dots\} \subseteq X$ is a set, which may or may not be infinite. $(p_i) = (p_i \mid i = 1, 2, \dots) = (p_1, p_2, \dots) \in X^{\mathbb{N}}$ is an infinite sequence from X . We might start counting from 0 instead of 1 on occasion, and if we need to start from another point, we make this explicit, using notation like $\{p_i\}_{i=k}^{\infty}$ and $(p_i)_{i=k}^{\infty}$. It is important to have these two notations, for example, if $p_i = (-1)^i$, then $\{p_i\} = \{-1, 1\}$ while $(p_i) = (-1, 1, -1, 1, \dots)$ and if $q_i = (-1)^{i+1}$, then $\{q_i\} = \{p_i\} = \{-1, 1\}$ while $(q_i) = (1, -1, 1, -1, \dots) = -(p_1)$.

Problem 3.1 (B:3.1:1 - A convergent sequence together with the point it approaches form a compact set.). Let X be a metric space.

- (a) Show that if (p_i) is a sequence from X with $p = \lim_{i \rightarrow \infty} p_i = p$, then $\{p\} \cup \{p_i\}$ is compact.
- (b) Deduce that for all $S \subseteq X$, S is closed iff for all compact $K \subseteq X$, $S \cap K$ is closed.

(a): Let $\mathcal{O}$ be an open cover of $K = \{p_i\} \cup \{p\}$. Let $O \in \mathcal{O}$ be such that $p \in O$. Since $p_i \rightarrow p$, $K \setminus O$ is finite. Choose finitely many sets $O_1, \dots, O_k \in \mathcal{O}$ so that $K \subseteq O_1 \cup \dots \cup O_k \cup O$, so that $\{O_1, \dots, O_k, O\} = \mathcal{O}' \subseteq \mathcal{O}$ is a finite subcover of K . Thus, we see K is compact.

(b): Assume $S \subseteq X$ is closed. Let K be compact, K is closed, since X is metric and hence Hausdorff. Hence $S \cap K$ is closed.

Conversely, assume $S \cap K$ is closed for all compact K . Let p be a limit point of S , so $p = \lim p_i$ where $p_i \in S$. From (a), $K = \{p_i\} \cup \{p\}$ is compact so by our assumption, $K \cap S$ is closed. As $p_i \in S \cap K$ for all i , we have $p \in S \cap K$ since $S \cap K$ is closed. We have shown that every limit point of S is in S , so S is closed.

Problem 3.2 (B:3.2.1 - Converse to Theorem 3.6 (a).). Let X be a metric space. Show that if every sequence (x_i) in X has a convergent subsequence, then X is compact.

Hint: You can use R:2.26 which was homework problem 2.7.

Note: The statement: Every sequence has a convergent subsequence is called *sequential compactness*. So this problem has you show that if X is metric, then: X is sequentially compact iff X is compact.

It is simple to show that, in metric spaces, compact $\implies$ sequentially compact. (Unlike with closure and sequential closure, a metric space is required here.)

In our current problem, we have sequential compactness. In order to apply R:2.26 we need a countable base for the topology. This we get using sequential compactness. For each $i = 1, 2, \dots$ build a finite open cover $\mathcal{O}_i$ of X using open sets of the form $N_{\frac{1}{i}}(x)$. Once we have these, let $\mathcal{O} = \bigcup \mathcal{O}_i$. $\mathcal{O}$ is a countable base. To see this, let U be open and $x \in U$. Let $N_\delta(x) \subseteq U$. Take i so that $\frac{1}{i} < \frac{\delta}{2}$. Since $\mathcal{O}_i$ is a cover, $x \in O \in \mathcal{O}_i$. Say $O = N_{\frac{1}{i}}(y)$. We see that $x \in O \subseteq U$ as required for $\mathcal{O}$ to be a base. This is just from the triangle inequality. Let $z \in O$, then

$$d(z, x) \leq d(z, y) + d(y, x) < \frac{1}{i} + \frac{1}{i} < \frac{\delta}{2} + \frac{\delta}{2} = \delta$$

So, $z \in N_\delta(x) \subseteq U$ and hence $z \in U$.

We must still show that the $\mathcal{O}_i$ exist. Take $x_1 \in X$ to be any point. If $N_{\frac{1}{1}}(x_1)$ does not cover X , then take $x_2 \in X \setminus N_{\frac{1}{1}}(x_1)$. Continue in this fashion choosing $x_m \in X \setminus \bigcup_{j=1}^{m-1} N_{\frac{1}{j}}(x_j)$. We must stop at some finite stage, at which point we have $\mathcal{O}_i$, otherwise, we get (x_m) where $d(x_m, x_k) \geq \frac{1}{i}$ for all $m \neq k$. But such a sequence could not have a convergent subsequence, contradicting sequential compactness.

Problem 3.3 (R:3.20). Let X be a metric space and (p_i) be a Cauchy sequence with a convergent subsequence. Show that (p_i) converges.

Suppose (p_{m_i}) converges to p . For each $\epsilon > 0$, there is N so that $d(p, p_{m_i}) < \epsilon/2$ for all $m_i > N_1$. Since (p_i) is a Cauchy sequence, there is N_2 so that for $i, j > N_2$, $d(p_i, p_j) < \epsilon/2$. Let $N = \max\{N_1, N_2\}$, let $i > N$, then $d(p_i, p) \leq d(p, p_{m_j}) + d(p_{m_j}, p_i) < \epsilon/2 + \epsilon/2 = \epsilon$ for any $m_j > N$. This shows that $p_i \rightarrow p$.

Problem 3.4 (R:3.22 - Baire Category Theorem). Suppose X is a complete metric space and $\{G_n\}$ is a set of dense open subsets. Show that $\bigcap G_n$ is dense.

We just construct a point in $\bigcap G_n$. In fact, let O be an arbitrary open set. There is $x_1 \in O \cap G_1$ since G_1 is dense. Since $O \cap G_1$ is open and we get $N_{\delta_1}(x_1) \subset O \cap G_1$ so that $\text{cl}(N_{\delta_1}(x_1)) \subset O \cap G_1$. Now let $x_2 \in N_{\delta_1}(x_1) \cap G_2$ and $N_{\delta_2}(x_2) \subset N_{\delta_1}(x_1) \cap G_2$ so that $\text{cl}(N_{\delta_2}(x_2)) \subset N_{\delta_1}(x_1) \cap G_2$. Just continue like this. We have F_i closed $F_i \subset G_i$ and $\text{diam}(F_i) \rightarrow 0$. Let $y_i \in F_i$, then $(\forall n, m > i) d(x_n, x_m) \leq \text{diam}(F_i)$. So (y_i) is a Cauchy sequence and by completeness, $y_i \rightarrow y$, we have that $y \in F_i$ for all i and so $y \in \bigcap F_i \subseteq O \cap (\bigcap G_i)$.

We have actually shown that $O \cap (\bigcap G_i) \neq \emptyset$ for all open O and hence that $\bigcap G_i$ is dense.

Note, we can easily find a sequence of open sets with descending diameters whose intersection is empty, e.g., just consider $O_i = (1/i, 1)$. Also, we can find a descending set of closed sets F_i with $\text{diam}(F_i) \not\rightarrow 0$ so that $\bigcap F_i = \emptyset$, consider $F_i = [i, \infty)$. So our descending sequence needs to consist of closed sets with decreasing diameters.

Problem 3.5 (Application of Baire Category).

(a) Show that $\mathbb{Q}$ is a countable union of closed sets.

(b) Show that $\mathbb{Q}$ is not a countable intersection of open sets.

(a): Let $\mathbb{Q} = \{q_i \mid i \in \mathbb{N}\}$, then $\mathbb{Q} = \bigcup_{i \in \mathbb{N}} \{q_i\}$. since each singleton $\{q_i\}$ is closed, $\mathbb{Q}$ is a union of countable many closed sets.

(b): Suppose $\mathbb{Q} = \bigcap U_i$ for U_i open. Since $\mathbb{Q} \subseteq U_i$ is dense, each U_i is open and dense. Let $V_i = U_i \setminus \{q_i\}$, V_i is open and still dense. We have $\bigcap V_i = \bigcap (U_i \cap \{q_i\}^c) = \bigcap U_i \cap \bigcap \{q_i\}^c = \mathbb{Q} \cap \mathbb{Q}^c = \emptyset$. But this contradicts the Baire category theorem.

Problem 3.6 (R:3:16(18)*). Fix $\alpha > 1$ and $x_1 > \sqrt{\alpha}$, define

$$x_{n+1} = \frac{1}{2} \left(x_n + \frac{\alpha}{x_n} \right)$$

a) Prove that (x_n) decreases monotonically and $\lim_{n \rightarrow \infty} x_n = \sqrt{\alpha}$.

A little argument and trick shows that $x_n^2 \geq \alpha$. The “trick” is to see that

$$\frac{x_n^2}{\alpha} + \frac{\alpha}{x_n^2} \geq 2$$

For this we need $x_n^4 + \alpha^2 > 2\alpha x_n^2$, but this is just $x_n^4 + \alpha^2 - 2\alpha x_n^2 = (x_n^2 - \alpha)^2 \geq 0$.

Now we can use this as follows:

$$x_{n+1}^2 = \frac{1}{4} \left(x_n^2 + 2\alpha + \frac{\alpha^2}{x_n^2} \right) = \frac{1}{4} \left(\frac{x_n^2}{\alpha} + 2 + \frac{\alpha}{x_n^2} \right) \cdot \alpha \geq \frac{1}{4} (2 + 2) \cdot \alpha = \alpha$$

Let's see now that $x_{n+1} \leq x_n$, this will use the above:

$$x_{n+1} = \frac{1}{2} x_n + \frac{1}{2} \left(\frac{\alpha}{x_n} \right) = \frac{1}{2} x_n + \frac{1}{2} \left(\frac{\alpha}{x_n^2} \right) x_n \leq \frac{1}{2} x_n + \frac{1}{2} x_n = x_n$$

We know $\gamma = \lim_{n \rightarrow \infty} x_n$ and by the standard limit laws

$$\gamma = \lim_{n \rightarrow \infty} x_{n+1} = \lim_{n \rightarrow \infty} \frac{1}{2} \left(x_n + \frac{\alpha}{x_n} \right) = \frac{1}{2} \left(\gamma + \frac{\alpha}{\gamma} \right)$$

Now we have:

$$2\gamma = \gamma + \frac{\alpha}{\gamma}$$

and hence $\gamma^2 = \alpha$, as desired.

b) Let $\varepsilon_n = x_n - \sqrt{\alpha}$ be the error at the n^{th} term, show that

$$\varepsilon_{n+1} = \frac{\varepsilon_n^2}{2x_n} < \frac{\varepsilon_n^2}{2\sqrt{\alpha}}$$

Setting $\beta = 2\sqrt{\alpha}$, gives

$$\varepsilon_{n+1} < \beta \left(\frac{\varepsilon_1}{\beta} \right)^{2^n}$$

$$\begin{aligned}
\varepsilon_{n+1} &= x_{n+1} - \sqrt{\alpha} = \frac{1}{2} \left(x_n + \frac{\alpha}{x_n} \right) - \sqrt{\alpha} \\
&= \frac{x_n^2 + \alpha - 2x_n\sqrt{\alpha}}{2x_n} = \frac{(\varepsilon_n + \sqrt{\alpha})^2 + \alpha - 2(\varepsilon_n + \sqrt{\alpha})\sqrt{\alpha}}{2x_n} \\
&= \frac{\varepsilon_n^2 + 2\varepsilon_n\sqrt{\alpha} + \alpha + \alpha - 2(\varepsilon_n + \sqrt{\alpha})\sqrt{\alpha}}{2x_n} = \frac{\varepsilon_n^2}{2x_n}
\end{aligned}$$

Since $x_n > \sqrt{\alpha}$ we do have

$$\varepsilon_{n+1} < \frac{\varepsilon_n^2}{2\sqrt{\alpha}} = \frac{\varepsilon_n^2}{\beta} = \beta \left(\frac{\varepsilon_n}{\beta} \right)^2 < \beta \left(\frac{\varepsilon_{n-1}}{\beta} \right)^2 \cdots < \beta \left(\frac{\varepsilon_{n-(n-1)}}{\beta} \right)^{2^n} = \beta \left(\frac{\varepsilon_1}{\beta} \right)^{2^n}$$

- c) Choose a number $\alpha > 3$ find a bound for how many terms are need to compute $\sqrt{\alpha}$ correct to 20 decimal places where x_1 is chosen minimally so that $x_1^2 > \alpha$. Prove your answer and do the computation. You might use [Python](#) or MATLAB.

This will vary, I'll take $\alpha = 145$ and hence $x_1 = 13$. $\beta = 2\sqrt{145} < 26$ and $\beta > 24$ and $\varepsilon_1 = 13 - \sqrt{145} < 1$ and so

$$\varepsilon_{n+1} < \beta \left(\frac{\varepsilon_1}{\beta} \right)^{2^n} < (26) \left(\frac{1}{24} \right)^{2^n}$$

So it would suffice to have n such that $(26)(24)^{-2^n} < 10^{-21}$, this is the same as $(1/26)(24)^{2^n} > 10^{21}$, a little playing shows $n = 4$ suffices, so take $n + 1 = 5$ terms. Testing with Python get

$$\begin{aligned}
x_1 &= 13 \\
\varepsilon_1 &= 0.958 \\
x_2 &= \mathbf{12.076923076923076923076923076923076923076923} \\
\varepsilon_2 &= 0.0353 \\
x_3 &= \mathbf{12.041646251837334639882410583047525722684958353748} \\
\varepsilon_3 &= 0.0000517 \\
x_4 &= \mathbf{12.041594578903165021043862247214042390984013945153} \\
\varepsilon_4 &= 1.11e - 10 \\
x_5 &= \mathbf{12.041594578792295480128751430185805358673288917241} \\
\varepsilon_5 &= 5.10e - 22
\end{aligned}$$

The actual value is:

$$\sqrt{145} = \mathbf{12.041594578792295480128241030378608052425352405054...}$$

- d) Replace the recursion above by

$$x_{n+1} = \frac{p-1}{p} x_n + \frac{\alpha}{p} x_n^{-p+1}$$

Discuss the behavior of (x_n) under suitable conditions. Don't bother trying to compute a recursive expression for ε_n in this case, but do prove your claims.

Assume $x_n^p > \alpha$, then

$$x_{n+1} = \left(\frac{p-1}{p}\right)x_n + \left(\frac{1}{p}\right)\left(\frac{\alpha}{x_n^p}\right)x_n \leq x_n$$

Now we must show that $x_{n+1} > \alpha^{1/p}$ given that $x_n > \alpha$. A little algebra gives:

$$\begin{aligned} x_{n+1} &= \frac{\alpha^{1/p}}{p} \left((p-1) \left(\frac{x_n}{\alpha^{1/p}} \right) + \left(\frac{\alpha^{1/p}}{x_n} \right)^{p-1} \right) \\ &= \frac{\alpha^{1/p}}{p} ((p-1)z^{-1} + z^{p-1}) \end{aligned}$$

It suffices to show that

$$(p-1)z^{-1} + z^{p-1} > p \quad (\dagger)$$

for then we get $x_{n+1} = \alpha^{1/p}(1/p)((p-1)z^{-1} + z^{p-1}) > \alpha^{1/p}$. In order to verify $(\dagger)$ it suffices to show $(p-1) + z^p > zp$ which is equivalent to $z^p - 1 > (z-1)p$ so we may show

$$\frac{z^p - 1}{z - 1} = z^{p-1} + z^{p-2} + \cdots + 1 > p \quad (\ddagger)$$

But as $z > 1$ we have $z^i > 1$ and so we have the sum of p -terms all ≥ 1 and where $p-1$ of them are > 1 , so $(\ddagger)$ holds.

So the series x_n is monotonic decreasing and bounded below by $\alpha^{1/p}$. Let $\gamma = \lim_{n \rightarrow \infty} x_n$ we have

$$\gamma = \lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} \left(\frac{p-1}{p}x_n + \frac{1}{p} \frac{\alpha}{x_n^{p-1}} \right) = \frac{p-1}{p}\gamma + \frac{1}{p} \frac{\alpha}{\gamma^{p-1}}$$

This gives $p\gamma^p = (p-1)\gamma^p + \alpha$ which is the same as $\gamma^p = \alpha$.

Problem 3.7 (B:3.7:1 - No boundary between convergent and divergent series.). The statement Rudin makes on p.63 that there is no “boundary” between convergent and divergent series can be made precise as follows: Given sequences (a_n) , (b_n) of positive real numbers, each having limit 0, let us say that “ (a_n) decays more rapidly than (b_n) ”, equivalently that “ (b_n) decays more slowly than (a_n) ” if $\lim_{n \rightarrow \infty} a_n/b_n = 0$, equivalently, $\lim_{n \rightarrow \infty} b_n/a_n = +\infty$. We note that if a sequence of positive terms has a convergent sum, so does every sequence of positive terms that decays more rapidly; and if a sequence of positive terms has a divergent sum, then so does every sequence of positive terms that decays more slowly.

Let us call (a_n) a “boundary sequence” if every sequence that decays more rapidly than (a_n) converges and every sequence that decays more slowly than (a_n) diverges. Rudin's **3:R11** and **3:R12** show respectively that given a divergent series $\sum a_n$, one can find a sequence (b_n) which decays more rapidly than (a_n) but such that $\sum b_n$ still diverges, and that given a convergent series $\sum a_n$, one can find a sequence (b_n) which decays more slowly than (a_n) but such that $\sum b_n$ still converges.

- (a) Deduce from the above facts that no series $\sum a_n$, whether divergent or convergent, can satisfy the definition given above for a “boundary sequence” (thus justifying Rudin’s claim). However, the constructions of **3:R11** and **3:R12** are unnecessarily complicated. Here is a simpler pair of constructions:
- (b) Suppose $\sum a_n$ is a divergent series of positive terms with $\lim_{n \rightarrow \infty} a_n = 0$. Show that there is an increasing sequence (A_n) of positive real numbers with $\lim_{n \rightarrow \infty} A_n = +\infty$ such that $A_{n+1} - A_n = a_n$ for all n . Now let $B_n = A_n^{1/2}$, and define $b_n = B_{n+1} - B_n$. Show that (b_n) is also a sequence of positive terms that approaches 0, and that it decays more rapidly than (a_n) , but that (B_n) still approaches $+\infty$, hence that $\sum b_n$ is still divergent. (Hint: Here and in part (b), the relation $B_n = A_n^{1/2}$ is more easily used in the form $A_n = B_n^2$).
- (c) Suppose $\sum a_n$ is a convergent series of positive terms. Show that there is a decreasing sequence (A_n) of positive real numbers with $\lim_{n \rightarrow \infty} A_n = 0$ such that $A_n - A_{n+1} = a_n$ for all n . Again let $B_n = A_n^{1/2}$ and now let $b_n = B_n - B_{n+1}$. Show that (B_n) also approaches 0, hence that $\sum b_n$ converges, but that (b_n) decays less rapidly than (a_n) .

That (a) follows from (b) and (c) is clear.

(b): Let $\sum a_n$ diverge with $a_n \rightarrow 0$. Let $A_n = \sum_{i=1}^{n-1} a_i$ so $A_{n+1} - A_n = a_n$. Since $\sum a_n$ diverges, $A_n \rightarrow \infty$. Let $B_n = A_n^{1/2}$ and $b_n = B_{n+1} - B_n$. We know $(B_{n+1} - B_n)(B_{n+1} + B_n) = B_{n+1}^2 - B_n^2 = A_{n+1} - A_n = a_n \rightarrow 0$ while $B_{n+1} + B_n \geq 2B_n \rightarrow \infty$ and so $b_n \rightarrow 0$. Note

$$\frac{a_n}{b_n} = \frac{A_{n+1} - A_n}{B_{n+1} - B_n} = \frac{B_{n+1}^2 - B_n^2}{B_{n+1} - B_n} = B_{n+1} + B_n \rightarrow \infty$$

So b_n decays more rapidly than a_n and yet $\sum b_n$ still diverges.

(c): Let $\sum a_n$ converges. Let $A_n = \sum_{i=n}^{\infty} a_i$ so $A_n - A_{n+1} = a_n$. Since $\sum a_n$ converges, $A_n \rightarrow 0$. Let $B_n = A_n^{1/2}$ and $b_n = B_n - B_{n+1}$. We know $(B_n - B_{n+1})(B_n + B_{n+1}) = B_n^2 - B_{n+1}^2 = A_n - A_{n+1} = a_n \rightarrow 0$ while $B_n + B_{n+1} \geq 2B_n \rightarrow 0$ and so $\sum b_n$ also converges. Note

$$\frac{a_n}{b_n} = \frac{A_n - A_{n+1}}{B_n - B_{n+1}} = \frac{B_n^2 - B_{n+1}^2}{B_n - B_{n+1}} = B_n + B_{n+1} \rightarrow 0$$

So b_n decays more slowly than a_n and yet $\sum b_n$ still converges.

Problem 3.8 (R:3.9). Find the radius of convergence and **region interval of convergence** of the following power series. If you take $z \in \mathbb{R}$, then the region of convergence is an interval, but if $z \in \mathbb{C}$, then this region is the open disk $N_R(0)$ together with part of its boundary. The complex region (the boundary) might be difficult to calculate, but you can calculate the interval of convergence for real values of z .

a) $\sum_n n^3 z^n$

$$\limsup \frac{(n+1)^3 |z|^{n+1}}{n^3 |z|^n} = |z| \text{ so this converges absolutely for } |z| < 1, \text{ that is, } R = 1.$$

b) $\sum_n \frac{2^n}{n!} z^n$

$$\frac{2^{n+1} |z|^{n+1}}{(n+1)!} \bigg/ \frac{2^n |z|^n}{n!} = \frac{2^{n+1}}{2^n} \cdot \frac{n!}{(n+1)!} \cdot |z| = \frac{2}{n+1} |z| \rightarrow 0. \text{ So the radius of convergence is } R = \infty.$$

- c) $\sum_n \frac{2^n}{n^2} z^n$
 $\frac{2^{n+1}|z|^{n+1}}{(n+1)^2} \bigg/ \frac{2^n|z|^n}{n^2} = \frac{2^{n+1}}{2^n} \cdot \frac{n^2}{(n+1)^2} \cdot |z| = 2 \cdot \frac{n^2}{(n+1)^2} \cdot |z| \rightarrow 2|z|$. This converges absolutely for $|z| < \frac{1}{2}$.
- d) $\sum_n \frac{n^3}{3^n} z^n$
 $\frac{(n+1)^3|z|^{n+1}}{3^{n+1}} \bigg/ \frac{n^3|z|^n}{3^n} = \frac{(n+1)^3}{n^3} \cdot \frac{1}{3} \cdot |z| = \frac{1}{3} \cdot \frac{(n+1)^3}{n^3} \cdot |z| \rightarrow \frac{1}{3}|z|$. This converges absolutely for $|z| < 3$.
- e) $\sum_n \frac{n^n}{n!} z^n$
 $\frac{n^{n+1}|z|^{n+1}}{(n+1)!} \bigg/ \frac{n^n|z|^n}{n!} = \frac{(n+1)^{n+1}}{n^n} \cdot \frac{n!}{(n+1)!} \cdot |z| = \left(\frac{n+1}{n}\right)^n |z| = \left(1 + \frac{1}{n}\right)^n |z| \rightarrow e|z|$. This converges absolutely for $|z| < \frac{1}{e}$.

It is interesting to see what happens at $z = \frac{1}{e}$ and $z = -\frac{1}{e}$. There is a nice approximation to $n!$ called Stirling's Formula, which is $n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$. So substituting this and $z = \frac{1}{e}$ gives $\sum_n \frac{n^n}{\sqrt{2\pi n}} \left(\frac{e}{n}\right)^n \left(\frac{1}{e}\right)^n = \sum_n \frac{1}{\sqrt{2\pi n}}$ which diverges by the p -test. At $z = -\frac{1}{e}$ we end up with $\frac{n^n}{n!} \left(\frac{1}{e}\right)^n \sim \frac{1}{\sqrt{2\pi n}} \rightarrow 0$ so we can use the alternating series test. Thus, the interval of convergence is $[-1/e, 1/e)$.

Problem 3.9 (B:3.13:1 - Radius of Convergence of sum and product series.). Let $\sum_n a_n$ and $\sum_n b_n$ be two series and let $\sum_n a_n + \sum_n b_n = \sum_n s_n$ and $(\sum_n a_n)(\sum_n b_n) = \sum_n p_n$. Recall that this means $s_n = a_n + b_n$ while $p_n = \sum_{i+j=n} a_i b_j = \sum_{i=0}^n a_i b_{n-i}$. Let R_a, R_b, R_s , and R_p be the radius of convergence of $\sum_n a_n, \sum_n b_n, \sum_n s_n$, and $\sum_n p_n$ respectively.

- (a) Show that $\min\{R_a, R_b\} \leq \min\{R_s, R_p\}$.
- (b) Deduce that if $R_a < R_b$, then $R_s = R_a$. What might happen if $R_a = R_b$?
- (c) Deduce that if $R_p < R_a$, then $R_b \leq R_p$.

(a) Suppose $r < \min\{R_a, R_b\}$, then we can find z_0 with $r < |z_0| < \min\{R_a, R_b\}$, then $\sum a_n z_0^n$ and $\sum b_n z_0^n$ are absolutely convergent. It is clear that $\sum s_n z_0^n$ converges, for $\sum p_n z_0^n$ apply 3.50 from Rudin. We know that if $\sum c_n z_0^n$ converges, then the radius of convergence of $\sum c_n z^n$ is $\geq |z_0|$. So we have $r < \min\{R_s, R_p\}$. This verifies that $\min\{R_a, R_b\} \leq \min\{R_s, R_p\}$.

(b): If $R_a < R_b$, then for any z with $R_a < |z| < R_b$ we know $\sum a_n z^n$ diverges and $\sum b_n z^n$ converges, so $\sum s_n z^n$ diverges. Thus we have $R_s \leq R_a$, but from (a), $R_a \leq R_s$, and so equality holds.

If $R_a = R_b$, we might be looking at $b_n = -a_n$ so that $\sum s_n z^n$ has radius of convergence $R = \infty$.

(c): If $R_p < R_a$, then $R_b \leq R_p < R_a$ by (a).

Problem 3.10 (B:3.14:2 - Which series have convergent rearrangements?). Find a simple and easy-to-check criterion for a series $\sum a_n$ to have a rearrangement $\sum a_{\sigma(n)}$ that converges, for some σ a permutation of $\mathbb{Z}^+$.

The criterion is: $\lim_{n \rightarrow \infty} a_n = 0$, and the sum of the positive terms and the sum of the negative terms are either both finite or both infinite.

To prove this, let $p_n = \max(a_n, 0)$ and $q_n = -\min(a_n, 0)$.

Necessity: Suppose a rearrangement $\sum a_{\sigma(n)}$ converges. By the n th-term test, $a_{\sigma(n)} \rightarrow 0$. Since σ is a bijection, the original sequence must also satisfy $a_n \rightarrow 0$. Furthermore, suppose for contradiction that $\sum p_n = P < \infty$ but $\sum q_n = \infty$. Because $\sum p_n$ converges absolutely, any rearrangement of the positive terms sums to P . Therefore, the partial sums of $\sum a_{\sigma(n)}$ are strictly bounded above by P , but driven to $-\infty$ by the diverging negative terms. Thus, the rearrangement diverges to $-\infty$, contradicting convergence. By symmetry, we cannot have $\sum q_n < \infty$ and $\sum p_n = \infty$ either. Thus, they must either both be finite or both be infinite.

Sufficiency: Suppose $a_n \rightarrow 0$. If both $\sum p_n$ and $\sum q_n$ are finite, the original series converges absolutely. In this case, every rearrangement converges, including the trivial one $\sigma(n) = n$. If both $\sum p_n$ and $\sum q_n$ are infinite, we can use the algorithm from Riemann's Rearrangement Theorem. We build the rearrangement by taking just enough positive terms to make the partial sum > 0 , then just enough negative terms to make the partial sum < 0 , and repeat. Because both sub-series sum to infinity, we never run out of terms. Because $a_n \rightarrow 0$, the magnitude of the "overshoot" past zero eventually vanishes, forcing the rearranged series to converge to 0.